

Biological Grounding and Parameterization of Artificial Ecosystem Morphogenesis

1. Introduction: Simulating Ecosystem Dynamics via Continuum-Agent Interactions

The development of a unified, GPU-computed ecosystem represents a sophisticated intersection of theoretical biology, complex systems modeling, and advanced computational rendering. In this architectural paradigm, a double-buffered simulation manager coordinates a read-modify-write cycle over a continuous spatial domain—defined as the "Biome"—while discrete, autonomous agents sample and alter this field via species-specific sensory-effector mappings, termed "Umwelts." Because the ultimate objective is an aesthetically driven art system displaying emergent morphogenesis rather than a strictly homeostatic, biologically closed loop, the parameterization must be carefully calibrated toward critical "edge-of-chaos" regimes.

To achieve qualitative plausibility and to generate rich, legible spatial structures—such as the reticulated transport networks of slime molds, the complex stigmergic architectures of termites, and the scale-free density waves of flocking birds—the abstract computational parameters governing these agents must be grounded in empirical biological data. By extracting relational orderings, dimensionless ratios, and allometric scaling principles from the literature concerning *Physarum polycephalum* (slime molds), *Macrotermes* (mound-building termites), and *Sturnus vulgaris* (flocking starlings), the simulation can accurately replicate the fundamental physical and biological rules that drive natural pattern formation.

This report provides an exhaustive synthesis of the biological literature required to parameterize this GPU-compute ecosystem. It systematically evaluates the sensory geometries, locomotor dynamics, metabolic scaling, thermodynamic interactions, and mortality mechanisms of the three chosen taxa. Furthermore, it conducts a rigorous evaluation of the current ten-channel Biome architecture, offering recommendations for structural adjustments and the addition of critical environmental layers that are biologically load-bearing, ultimately bridging the gap between biological reality and computational aesthetics.

2. Evaluation and Expansion of the Environmental Biome

The foundational environment of the simulation relies on a double-buffered multi-channel texture array that acts as the shared chemical and physical substrate. The current implementation features ten scalar field layers: Nutrient, three Pheromone layers, Oxygen, Temperature, Waste, Permeability, and two advection velocity vectors (Flow_X, Flow_Y). While this provides a robust baseline for basic reaction-diffusion and advection mechanics, true biological morphogenesis is fundamentally contingent on several other environmental forces.

An evaluation of candidate layers is necessary to ensure the environment supports the specific behavioral quirks of the chosen taxa.

2.1 The Primacy of Humidity and Moisture Gradients

An analysis of the physiological and behavioral literature indicates that moisture is arguably the most glaring omission from the current biome architecture. For both *Physarum polycephalum* and *Macrotermes*, moisture gradients are physically more determinative than temperature gradients.

In the context of termite construction, the primary cue for stigmergic building is not a highly specific chemical cement, but rather local evaporation flux.¹ Evaporation is governed by the gradient of humidity. Termites naturally dig along the humidity gradient, moving toward regions of higher moisture to prevent desiccation, and they deposit building materials at local maxima of evaporation.¹ Similarly, the plasmodium of *Physarum polycephalum* is essentially a complex hydraulic network of cytoplasm. Desiccation serves as a primary trigger for senescence and the morphological phase transition into dormancy.²

The addition of a dedicated Humidity layer is strongly recommended. This layer should possess a high diffusion rate and be actively consumed by agents to simulate hydration. Furthermore, the Temperature channel must negatively couple with the Humidity channel to mathematically simulate evaporation. In the termite Umwelt, the deposition of Permeability should be intricately mapped to regions where Humidity gradients are steepest, thereby simulating the high evaporation flux that guides natural architectural efforts.

2.2 Topography and Substrate Curvature

The current Permeability channel operates as a flat, binary representation of solid versus open matter. However, true mound morphogenesis requires three-dimensional topographic cues. Empirical studies of *de novo* termite mound construction reveal that deposition is explicitly triggered by convex curvature, such as the tips of pre-existing pillars or the elevated edges of the substrate, whereas concave regions preferentially attract excavation activity.¹

A flat permeability map cannot natively compute curvature. Therefore, it is highly recommended to conceptualize the Permeability layer as a Topography or Height map, where the scalar value of a pixel represents the vertical height of the soil substrate. The simulation manager can easily compute the Laplacian of this Height map via a standard convolution matrix in a compute shader to derive the local surface curvature. The termite Umwelt can then be programmed to read this curvature channel: highly positive values (convexity) will trigger the deposition of material, while highly negative values (concavity) will trigger excavation.¹ This architectural modification grounds the simulation perfectly in recent empirical discoveries regarding stigmergic construction, replacing the biologically flawed reliance on a localized "cement pheromone."

2.3 Carbon Dioxide and Metabolic Hypoxia

Termite mounds are renowned for maintaining extreme carbon dioxide gradients, often reaching concentrations of up to 8%.⁴ The diurnal, lung-like convection of the mound explicitly

serves to flush this accumulated CO₂ while drawing in fresh air.⁵ Furthermore, termite microbiomes feature methanogenic and hydrogenotrophic bacteria, resulting in significant emissions of trace gases.⁴

Despite its biological significance, adding a separate CO₂ channel in a generalized two-dimensional simulation may be computationally redundant if its dynamics simply mirror the inverse of the Oxygen channel. Because Oxygen is already consumed metabolically by the agents, its localized absence inherently defines a hypoxic and hypercapnic zone. To optimize GPU memory resources—which are strictly constrained by texture channel counts—a discrete CO₂ layer is unnecessary. Instead, the inverse magnitude of the Oxygen channel can be utilized directly by the Umwelt mappings to represent suffocation, trigger avoidance behaviors, and stimulate convective flow responses.

2.4 External Drivers: Light and Diurnal Convection

The biological behaviors of the simulated taxa are heavily influenced by phototaxis and diurnal rhythms. *Physarum polycephalum* is naturally photophobic, avoiding light to prevent the desiccation of its plasmodium.⁷ Flocking birds, such as starlings, are strictly diurnal. Most critically, the convective ventilation of termite mounds is entirely driven by diurnal sunlight heating the exterior flutes of the structure.⁶

However, dedicating an entire scalar field layer to Light is highly inefficient. Light in this context does not behave as a diffusing chemical. Instead, Light should be implemented as a global uniform variable or a purely procedural mathematical gradient computed dynamically within the shaders. A sweeping, periodic gradient—simulating a sun passing overhead—can dynamically inject heat into the Temperature field based on the substrate's topography. This provides the external thermodynamic forcing necessary to drive the mound's cyclical convection, pushing the ecosystem away from static equilibrium and transforming it into a continuously driven art system.

2.5 Mass Conservation via Soil Load State

In the current implementation, termites execute writes to the Permeability channel to build or erode their environment. However, if a termite agent merely erodes permeability, that material must be accounted for mathematically. If it simply vanishes, mass is not conserved, and macroscopic mound structures cannot emerge or grow over time.

Rather than dedicating a full Biome layer to represent the moving soil load, this functionality should be incorporated directly into the agent data structure. A single floating-point value or boolean flag—`carryingSoil`—should be added to the 20-byte agent struct. When a termite excavates (triggered by reading high permeability or concave curvature), it decrements the Permeability field and toggles the `carryingSoil` state to true. When it subsequently encounters a deposition zone (triggered by high evaporation or convex curvature), it increments the

Permeability field and toggles the state to false. This computationally lightweight solution conserves mass and allows for true structural reorganization.

2.6 Wind Forcing vs. Internal Convection

Direct in-situ measurements of *Macrotermes* mounds explicitly rule out steady external wind as the primary driver of internal ventilation; rather, the process is overwhelmingly dominated by internal thermal convection driven by transient temperature differentials.⁶ Therefore, an independent directional wind forcing layer is not required. The current implementation, wherein Temperature gradients generate the advection velocity vectors Flow_X and Flow_Y, is mathematically robust and biologically accurate. It allows the fluid dynamics of the system to remain self-contained and elegantly coupled to the metabolic and solar heat inputs.

3. Sensory Geometry and Locomotor Dynamics of *Physarum polycephalum*

Physarum polycephalum is a macroscopic, multinucleate amoeboid organism renowned for its ability to form highly optimized, fault-tolerant transport networks through a process of self-organized foraging. The computational model developed by Jeff Jones (2010) accurately captures this morphogenesis using simplistic paracrine chemotaxis behaviors within a dissipative environment.⁷ Grounding this behavior requires a detailed mapping of biological observations to the exposed parameters of the simulation.

3.1 Chemotactic Preferences and Umwelt Mapping

In nature, the plasmodium responds to complex chemical gradients. Research using solid phase micro-extraction coupled with gas chromatography-mass spectrometry has identified numerous volatile organic compounds (VOCs) that attract or repel *Physarum*.¹¹ The slime mold exhibits strong positive chemotaxis toward attractants with a strict hierarchy of preference, such as farnesene, beta-myrcene, and tridecane.¹¹ Conversely, it exhibits negative chemotaxis (avoidance) when exposed to repellents like nonanal and benzaldehyde.¹¹ When an attractant and a repellent are mixed, the repellent suppresses the signal, demonstrating a hierarchy of inhibition over attraction.¹¹

In the PhysarumSim Umwelt mapping, the reads array must reflect this sensitivity. The organism must sample its own Pheromone_0 channel with a strong positive weight to simulate the autocatalytic paracrine feedback that drives network tube formation.⁷ The weights for the Boid and Termite pheromones should act as mild attractants if designated as food sources, or strong repellents if designated as competitors. The chemotaxis effect collapses these weighted inputs into the perception texture, guiding the agent's steering.

3.2 Sensor Angles, Turn Rates, and Spatial Dynamics

In the agent-based approximation of *Physarum*, the sensory apparatus is strictly defined by three geometric parameters: senseAngle (SA), senseDistance (SO), and turnAngle (RA).⁷ The agent projects three sensors forward (left, center, right) to sample the underlying chemical

field.

To achieve the labyrinthine, reticulated, and branching patterns characteristic of *Physarum*, the simulation must operate near specific critical thresholds. Pattern formation relies heavily on the hysteresis between the step size (moveSpeed) and the sensory reach. Morphogenesis occurs when the senseDistance is moderately larger than the moveSpeed, allowing the agent to anticipate gradients before physically traversing them.¹² The turnAngle must be relatively sharp (e.g., 30 to 45 degrees) to allow for the abrupt bifurcations that characterize real plasmodial networks.¹⁴ If the continuous trail diffuseRate is too high, the chemical field blurs excessively, causing the network to collapse into a unified, featureless blob. Conversely, if the decayRate is too fast, the trails disconnect entirely. Tuning these parameters to a delicate balance allows for the spontaneous formation and autonomous maintenance of complex network lacunae.⁷

3.3 Shuttle Streaming and Oscillatory Locomotion

A fundamental biological reality of *Physarum* is that its movement is not continuous. It is driven by periodic cytoplasmic streaming, termed "shuttle streaming," which is generated by the rhythmic contractions of the intracellular actomyosin cytoskeleton.³ This flow oscillates back and forth with an intrinsic period of approximately 1 to 2 minutes.¹⁷ Furthermore, empirical studies have identified longer, size-independent morphological oscillations with a period of 20 ± 10 minutes that dictate large-scale branching, front expansion, and overall shape formation.¹⁸ Overall, *Physarum* migrates at relatively slow speeds, typically covering a few millimeters per hour, though it can reach maximum speeds of up to 4 centimeters per hour under optimal conditions.³ Computationally, the agent moveSpeed must be calibrated as a very small fraction of the spatial domain per step, establishing it as the slowest organism in the ecosystem. To achieve true biological fidelity and prevent the simulated plasmodium from settling into stagnant equilibrium, the simulation should incorporate temporal oscillations into the moveSpeed and depositAmount parameters. Modulating these values with a dual-frequency sine wave—mimicking both the fast 1-2 minute peristaltic wave and the slow 20-minute morphological wave—will introduce naturalistic pulsing, peristalsis, and branching dynamics into the GPU ecosystem.

4. Stigmergic Construction and Swarm Dynamics of *Macrotermes*

The termite simulation (TermiteSim) is tasked with modeling both foraging trails and large-scale mound construction. Grounding these parameters requires dismantling classical assumptions regarding insect stigmergy and integrating modern empirical findings on evaporation flux, probability-based aggregation, and vibrational acoustic communication.

4.1 Foraging Trails and Pheromone Persistence

Termites utilize a complex cocktail of volatile and non-volatile pheromones to coordinate foraging and recruitment.¹⁹ A critical parameter for the continuous Biome layer is the decay rate of this chemical trail. Empirical measurements demonstrate that termite trail pheromones

retain detectable activity for several hours, with an estimated half-life of approximately 2.0 to 2.5 hours.²¹

While simple ant-colony optimization models often assume rapid evaporation to facilitate quick path convergence, a slightly extended persistence is biologically accurate for termites. Slower evaporation (e.g., modeled mathematically as a 5% evaporation rate per discrete time step in some simulations) allows the trail to remain accessible long enough for termite colonies to follow it without clustering excessively in one spot, leading to more stable population dynamics.²² In the context of the GPU ecosystem, the `decayRate` of `Pheromone_2` (the termite scent) must be set significantly lower than the decay rate of the *Physarum* trail. This temporal persistence allows established termite galleries to act as medium-term external memories, guiding subsequent generations of workers.²³

4.2 The Mechanics of Excavation and Aggregation

Historically, computational models of termite mound construction relied heavily on the "cement pheromone" hypothesis proposed by Grassé (1959). This theory posited that termites add a specific chemical to deposited soil pellets, which acts as a powerful attractant, triggering a positive feedback loop of further deposition by other workers.²⁵ However, extensive behavioral studies have fundamentally challenged this classical view, demonstrating that such a pheromone is not the primary organizing cue.¹

De novo construction is primarily organized around the act of excavation rather than isolated deposition.²⁶ Termites concentrate their digging activity at a few key sites, effectively creating large quarries. Crucially, their behavior is mediated by a mechanism of aggregation: an individual termite is significantly more likely to join an ongoing excavation site as the local density of other working termites increases.²⁶ Statistical modeling of this behavior reveals that the probability of joining an excavation is driven by an extremely strong positive influence of the number of termites present (aggregation coefficient: 4.12), completely independent of chemical cues.²⁶

In the TermiteSim, the rules governing the erosion of the Permeability layer must reflect this. The probability of an agent excavating material should be positively correlated with a high local agent density within its perception radius.

4.3 Deposition Rules and Evaporation Flux

Furthermore, termites do not carry soil over long distances; rather, they deposit it immediately along the edges of the active quarries, meaning deposition is heavily constrained to the boundaries of active digging sites.²⁶ The specific trigger for dropping a soil pellet is strongly modulated by local surface curvature, which termites sense indirectly via substrate evaporation flux.¹

The underlying physics dictates that the evaporation flux of a wet substrate is directly proportional to its local curvature. Maximum evaporation occurs at highly convex regions, such as the tips of pre-existing pillars or the upper edges of walls. This localized high-humidity gradient (which can vary from 10% to 100% across a distance no larger than a termite's body

length) intensely attracts pellet deposition.¹ Conversely, concave regions attract digging activity, as termites naturally excavate along the gradient toward the most humid regions of the substrate.¹

Therefore, the depositProbability in the TermiteSim must not be driven by a local "cement" pheromone channel. Instead, the deposition logic should evaluate the Laplacian curvature of the Topography/Permeability channel. The probability of building (depositing soil from the carryingSoil state) must be overwhelmingly positive at convex geometries, creating a highly robust, self-sustaining positive feedback loop that mimics true architectural morphogenesis without relying on arbitrary chemical markers.

4.4 Vibrational Alarm Communication and the "Firing" Mechanism

The current system architecture includes a unique "firing" mechanism driven by external neural recordings, which dynamically increases the moveSpeed and depositAmount of specific termite agents. Biologically, this computational feature maps perfectly to the termite vibrational alarm response.

When disturbed by predators or severe environmental shifts, soldier termites engage in rhythmic head-banging against the substrate, producing high-frequency vibrational alarm signals.²⁰ These mechanical waves propagate through the subterranean galleries at a remarkably high velocity of approximately 1.3 meters per second.²⁰ This acoustic transmission is drastically faster than chemical diffusion or physical locomotion, allowing the alarm to spread rapidly across the colony.²⁰

Nestmates are exquisitely sensitive to these vibrations, possessing a behavioral threshold amplitude of just 0.012 m/s^2 .²⁰ The reception of this signal induces a polyethic response: workers retreat rapidly, while soldiers are recruited to the source and begin drumming themselves, socially re-amplifying the signal.²⁰ While lower levels of disturbance are communicated via these vibratory signals, extreme disturbances trigger the release of volatile alarm pheromones (e.g., alpha-pinene and myrcene), which induce panicked running and significantly higher speeds.²⁹

In the simulation, the neural-triggered firingSpeedMul should represent a massive, instantaneous acceleration, simulating the acoustic wave propagation of the alarm state. A firing termite should move at an order of magnitude faster than its base moveSpeed, while simultaneously depositing a high concentration of a specialized "alarm" value into the Pheromone channel, triggering an autocatalytic panic response in nearby agents.

5. Topological Flocking Metrics of *Sturnus vulgaris*

The BoidSim utilizes a GPU spatial-hash algorithm to compute flocking dynamics. To capture the mesmerizing, scale-free murmuration density waves characteristic of starlings (*Sturnus vulgaris*), the interaction topology must be strictly defined according to advanced empirical metrics.

5.1 Metric vs. Topological Interaction Ranges

Classical Reynolds boids and early self-propelled particle models traditionally utilized metric

interaction ranges, where an individual agent reacts to all neighbors within a specified physical radius. However, exhaustive empirical field studies utilizing stereo-photography of starling flocks containing thousands of birds have conclusively proven that avian interactions rely on topological distance rather than metric distance.³²

A bird does not interact with all flock-mates within a certain volume; instead, it interacts with a fixed number of nearest neighbors, entirely irrespective of the spatial density of the flock.³²

Theoretical models have shown that topological models are significantly more stable than metric ones, particularly when facing predatory disruption or executing complex maneuvers.³²

5.2 The Specificity of Topological Rulings

The biological literature provides precise values for these topological interactions. For the behaviors of alignment (matching velocity vectors) and cohesion (steering toward the center of mass), starlings consistently adjust their movement based on their 6 to 7 nearest topological neighbors.³⁴ This specific ruling is crucial for maintaining flock cohesion and is responsible for the phenomenon known as "scale-free correlation." In a true murmuration, the correlation length of velocity fluctuations among group members grows linearly with the total size of the flock, meaning that a turn initiated by one bird can seamlessly propagate through thousands of individuals.³⁵

However, recent computational analyses investigating internal group dynamics have revealed that the mechanism for separation (collision avoidance) operates differently. Escape behavior and collision prevention are most effective when a bird avoids only its single closest neighbor, rather than compromising its trajectory by attempting to avoid 6 or 7 neighbors simultaneously.³⁴ Computational models, such as the StarDisplay system, confirm that the internal stability and group-level diffusion of a flock resemble empirical data much more closely when individuals restrict avoidance maneuvers strictly to their 1st nearest neighbor.³⁴

In mapping this to the BoidSim spatial-hash, the parameterization must execute a strict topological sort. The separationWeight must be computationally restricted to evaluate only the

$N = 1$ closest agent, enforcing a tight, immediate repulsion. Conversely, the

alignmentWeight and cohesionWeight must average their vectors over the $N = 6$ or

$N = 7$ closest agents.

5.3 Speed Controls and Temporal Kinetics

Starlings are highly agile aviators, flying at cruising speeds of approximately 10 m/s (roughly 40 km/h), which equates to covering approximately 100 body lengths per second.³⁶ The internal dynamics and the scale-free nature of the flock are highly sensitive to speed control. If speed is controlled too rigidly (forcing all birds to fly at exactly the same velocity), the correlation length is reduced, saturating at a fixed value and destroying the scale-free coordination.³⁵

In the simulation, the maxSpeed parameter for Boids must be vastly higher than the base moveSpeed of *Physarum* or Termites. Given the scale-independence of the system, if a Termite's base speed is defined as 1 unit per step, the Boid maxSpeed must be situated on the

order of 10^2 to 10^3 units per step. Furthermore, the maxForce parameter (which dictates turning acceleration) must be finely balanced. If maxForce is too low, the flock behaves sluggishly, resembling a viscous fluid rather than a highly reactive biological entity. If maxForce is too high, the flock is prone to chaotic fragmentation. Establishing this clear hierarchy of temporal kinetics across the ecosystem is essential for the aesthetic legibility of the three distinct species.

6. Metabolic Scaling, Thermodynamics, and Nutrient Cycling

The Biome layers for Oxygen, Temperature, Waste, and Nutrient represent the thermodynamic and metabolic currency of the simulation. To correctly configure the Umwelt mappings and the global cross-field couplings, the system must adhere to the mass-specific metabolic rates and thermodynamic rules of the target organisms.

6.1 Mass-Specific Metabolic Rates and Oxygen Consumption

Organisms continuously consume Oxygen and deposit Temperature (representing metabolic heat) into the Biome. The relative rates among the three species are highly stratified, following well-documented principles of metabolic allometry.

Sturnus vulgaris (Boids): Avian flight is arguably the most metabolically demanding locomotor strategy in the animal kingdom, requiring extraordinary physiological adaptations. The flight muscles of hummingbirds and small passeriform birds possess exceptionally high oxygen demand and maximum enzyme activities.³⁶ For starlings specifically, empirical measurements of oxygen consumption during horizontal flapping flight indicate a mass-specific power input of approximately 28.1 W/kg.³⁷ This flight metabolic rate is nearly a ten-fold increase over their basal resting metabolic rate.³⁸ Furthermore, their rapid respiration contributes to significant water loss.³⁸

Consequently, the oxygenConsumption and metabolicHeat parameters in the Boid Umwelt must be definitively the highest in the simulation. This enforces a dynamic spatial mechanism where dense murmuration clusters act as highly mobile "vacuums" that instantly strip Oxygen from the Biome, while simultaneously acting as intense, localized sources of Temperature deposition.

Macrotermes (Termites): Termite metabolic rates follow a pattern of negative allometric scaling. While individual mass-specific metabolic rate scales predictably, the collective metabolic rate of an entire colony often exhibits hypometric scaling (e.g., scaling with colony mass to the power of 0.75), suggesting a significant "group effect" where per-capita energy expenditure decreases in larger aggregates.³⁹ Nevertheless, the absolute mass-specific oxygen consumption remains substantial; individual termites consume between 1000 and

2000 μ l O₂/g/h, depending on their developmental stage and caste.⁴¹

Within the confined space of a mound, the collective metabolism of millions of termites generates intense heat and severely alters the atmospheric composition.⁴ In the Termite

Umwelt, the oxygenConsumption parameter should be set to a moderate level per agent. However, due to their stigmergic swarming behavior, their intense spatial clustering will result in severe local hypoxia and dramatic heat accumulation, driving the physical convection of the system.

Physarum polycephalum: As an amoeboid organism, *Physarum* exhibits vastly lower metabolic demands. Respiration rates are closely tied to its developmental phase, decreasing

from roughly $62 \mu\text{I O}_2/\text{mg protein/hr}$ during periods of active growth and streaming, down to

$17.5 \mu\text{I O}_2/\text{mg protein/hr}$ during the transition into spherulation (dormancy).² Furthermore, empirical studies have shown that when *Physarum* is exposed to simultaneous electric and magnetic fields—an analogue to the simulation's external "firing" neural inputs—it experiences depressed oxygen consumption and a significantly lengthened mitotic cycle.⁴²

The oxygenConsumption and metabolicHeat parameters for *Physarum* should be configured at the absolute baseline minimums. This metabolic frugality allows the slime mold to survive and slowly propagate in the severely hypoxic zones left in the wake of hyper-metabolic Boid flocks or dense Termite swarms, occupying a distinct ecological niche within the chemical field.

6.2 The Thermodynamics of Decomposition

The global cross-field coupling parameter wasteToNutrientRate is intended to simulate the decomposition of organic matter. In terrestrial ecosystems, decomposition is an entirely temperature-dependent process. This relationship is strictly quantified in biogeochemistry by the Q_{10} temperature coefficient.

The Q_{10} coefficient represents the factor by which the rate of a chemical or biological reaction increases for every 10°C rise in temperature.⁴⁴ For the decomposition of soil organic matter and biological waste, the median Q_{10} value is consistently measured at approximately 2.74, with the vast majority of ecological data falling strictly between 2.0 and 3.0.⁴⁵

In the computational Biome, the rate at which the Waste layer is converted into the Nutrient layer must not be modeled as a linear function. Instead, it must scale exponentially with the local value of the Temperature field. The conversion equation per pixel should be modeled mathematically as:

$$R_{decomp} = R_{base} \times Q_{10}^{\frac{T - T_{ref}}{10}}$$

Where the coefficient $Q_{10} \approx$, T is the local Biome temperature, and T_{ref} is an arbitrary baseline temperature. This non-linear dynamic ensures that areas of the Biome intensely heated by Boid flocks or Termite swarms experience rapid nutrient cycling. This burst of Nutrient directly fuels the chemotactic foraging of *Physarum*, creating a tightly coupled, self-organizing trophic cascade driven entirely by localized thermodynamic events.

6.3 Solar Forcing and Diurnal Mound Ventilation

The simulation utilizes the Temperature channel to generate physical advection velocities (Flow_X, Flow_Y). This mechanism maps elegantly to the actual physics of termite mounds, although recent biological discoveries provide a crucial nuance regarding its execution. For decades, scientific consensus assumed that termite mounds were ventilated primarily by internal metabolic heat rising or by steady external winds blowing across the structures.⁶ However, direct in-situ measurements using custom-designed thermal flow sensors have overthrown this paradigm. The mounds of *Macrotermes* and *Odontotermes* are actually ventilated by diurnal ambient temperature oscillations driven entirely by sunlight.⁵ The architecture of the mound features thin outer conduits (flutes) surrounding a massive central chimney. During the day, the flutes heat up rapidly relative to the insulated interior, pushing air up the exterior walls and forcing it down the central chimney in a closed convection cell.⁶ At night, the exterior cools rapidly while the interior retains heat, completely reversing the convective flow, pulling air up from the nest.⁵ The mound essentially acts as a solar-powered lung, taking one deep breath per day to flush out CO₂ and regulate interior microclimates. Computationally, the temperatureToFlowStrength parameter is highly realistic. However, to generate the true, cyclic morphogenesis of the mound, the system requires periodic thermal forcing. As recommended previously, introducing a sweeping, planar thermal gradient across the Temperature field—simulating the diurnal cycle of the sun—will force the internal convective flows to reverse periodically. This shifting fluid dynamics will dynamically sculpt the Permeability layer as Termites continuously respond and adapt to the fluctuating evaporation gradients.

7. Population Dynamics, Mortality, and Aesthetic Ecosystem Cycles

Although the ultimate goal of the system is an expressive, driven art installation rather than a homeostatic biological simulation, implementing realistic birth and death mechanisms acts as a profound aesthetic driver. Mortality drivers prevent the simulation from saturating the spatial domain, producing beautiful rhythmic blooms and dramatic ecological collapses.

7.1 Hypoxia, Desiccation, and Phase Transitions

Each species must respond uniquely to hostile shifts in the Biome, specifically the depletion of Oxygen and Nutrient channels.

1. **Termites:** Termites require strict moisture control and are highly sensitive to hypoxia and desiccation. Their deathThresholdOxygen should be calibrated relatively high. When local Oxygen is depleted due to intense over-aggregation, mass die-offs occur. These events deposit significant amounts of organic material into the Waste channel. Under the residual high Temperature of the collapsed colony, this Waste rapidly decomposes into Nutrient, hyper-fertilizing the local Biome and preparing the canvas for the next successional wave of organisms.⁵
2. **Physarum:** The slime mold possesses a highly unique survival strategy. When exposed to

adverse conditions—such as severe starvation, hypoxia, or desiccation—*Physarum* does not immediately undergo necrosis. Instead, it executes a morphological phase transition, cleaving its plasmodium into hard-walled, highly resilient dormant units called spherules (or microcysts), drastically reducing its respiration rate.² In the simulation, the *Physarum* deathThresholdOxygen should be set extremely low. Upon crossing this threshold, rather than dying and depositing corpseWasteAmount, the agents should halt their moveSpeed and cease all Pheromone deposition. They become static "spores," locked in place until the local Nutrient or Oxygen levels are replenished by shifting environmental flows, at which point they dramatically reactivate.

3. **Boids:** Due to their extreme metabolic requirements, flocking agents are highly vulnerable to both starvation and hypoxia. Flocks that strip a spatial region of its Oxygen will suffer rapid, catastrophic attrition. Their corpseWasteAmount should be exceptionally high, representing a massive caloric dump into the ecosystem upon death, fueling localized blooms of decomposition.³⁸

7.2 Parameter Regimes for Critical Morphogenesis

To achieve legible, expressive morphogenesis, the computational parameters must be situated near critical phase transitions—the "edge-of-chaos" regime where order and randomness interact to produce complexity.

- **Physarum Reticulation:** The signature reticulated networks and cellular lacunae of the slime mold only emerge when the senseDistance outpaces the moveSpeed, combined with a sharp turnAngle (RA). If the continuous trail diffuseRate outpaces the agent's ability to reinforce the path, the network dissolves. Tuning these parameters to a critical balance allows the agents to autonomously maintain the structural integrity of the network while continually exploring new territory.⁷
- **Termite Mound Emergence:** Mound morphogenesis requires exploiting the hysteresis between excavation and building. Mounds will only emerge if the preferredPermeability actively restricts movement, forcing the swarm to aggregate at physical boundaries. This requires a remarkably slow decayRate for termite trail pheromones to maintain spatial memory, combined with a high computational threshold for Permeability erosion to simulate the physical resistance of packed soil.²⁶
- **Flocking Density Waves:** The mesmerizing, scale-free dynamics of the flock only emerge at high kinematic speeds (maxSpeed) combined with strict topological rules. If maxForce (the limit on acceleration) is too low, the flock fails to execute the sharp, synchronized turns that define murmurations. If maxForce is high, but the alignment is metric rather than topological, the flock inevitably shatters into chaotic, disconnected fragments.

8. Synthesized Parameterization Guide

To ensure the Unity GPU-compute ecosystem functions as a unified, qualitatively accurate model of biological morphogenesis, the dimensionless parameters must strictly adhere to the relative orderings established in this research. The following tables synthesize the parameter space, mapping biological reality to the exposed computational variables.

Table 1: Temporal Kinetics and Lifespan Parameters

Parameter Concept	Physarum	Termite	Boids	Biological Rationale
moveSpeed	$1x$ (Baseline)	$10x$	$1000x$	Slime mold flows at mm/h ³ ; Termites traverse at cm/s; Starlings cruise at ~10 m/s. ³⁶
trailDecayRate	Moderate	Very Slow	N/A	Termite orientation trails persist for 2.5 hours. ²¹ Physarum trails require constant physical refresh.
oscillationPeriod	Fast (1-2m)	N/A	N/A	Peristaltic shuttle streaming is physically required for branching and network generation. ¹⁷
firingSpeedM ul	Decelerates	Massive ($+50x$)	N/A	Termite vibrational alarm propagation reaches 1.3 m/s, causing a vast kinetic amplification. ²⁰

Table 2: Metabolic and Thermodynamic Couplings

Parameter Concept	Relative Setting / Ratio	Biological Rationale
oxygenConsumption	Boid ($100x$) > Termite ($10x$) > Physarum ($1x$)	Extreme avian flight metabolism (28 W/kg) ³⁷ versus the near-dormant states of amoeboid spherules. ²
metabolicHeat	Boid ($10x$) > Termite ($5x$) > Physarum ($1x$)	High avian heat dissipation versus the localized thermal mass generated by dense termite swarm clustering.
wasteToNutrientRate	Exponential to Temp ($Q_{10} \approx$)	The decomposition of soil organic matter scales exponentially with temperature. ⁴⁵
temperatureToFlow	High	Diurnal solar thermal oscillations drive the deep convective ventilation of termite mounds. ⁶

Table 3: Topological Sensory Interactions (Umwelt)

Sensory Metric	Boids	Termites	Physarum
Avoidance	$N = 1$ Nearest Neighbor	Permeability Obstacles	Repellent VOCs / Competitor Trails
Alignment	$N = 6$ to 7 Neighbors	Topographic / Trail Gradients	Autocatalytic Pheromone Gradients
Cohesion	$N = 6$ to 7 Neighbors	High Density Excavation Sites	Dense Cytoplasmic Nodes

Stigmergy Logic	N/A	Evaporation / Convex Curvature	Paracrine feedback deposition
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9. Systemic Integration and Aesthetic Implications

The transition of a multi-agent particle system from a generic, chaotic simulation into a highly expressive, biologically grounded engine of morphogenesis requires absolute mathematical discipline regarding its kinetic, sensory, and thermodynamic properties.

This rigorous analysis highlights three critical paradigm shifts necessary for the successful execution of the ecosystem:

1. **The Absolute Dominance of Topological Flocking:** Boid dynamics must abandon the simplistic geometries of purely metric radii. The generation of beautiful, scale-free density waves relies strictly on averaging alignment across exactly 6 to 7 topological neighbors, while avoidance forces must be localized exclusively to the single closest neighbor.
2. **Topography Outcompeting Chemistry in Stigmergy:** The classical assumption of a highly specific "cement pheromone" coordinating termite construction is biologically obsolete. Termite Umwelt mappings must read the local Laplacian curvature of the Permeability layer, dropping soil at convex geometries driven by evaporation flux, and aggregating their excavation efforts at concave sites based on agent density.
3. **Thermodynamic and Solar Forcing:** To prevent the ecosystem from falling into a visually stagnant, static equilibrium, the conversion of Waste to Nutrient must be

modeled via the Q_{10} exponential temperature curve, and the advection of fluids must be driven by cyclic, diurnal thermal injections simulating the passage of light.

By meticulously integrating the allometric scaling of metabolic rates, the drastically stratified locomotor velocities, and the unique mortality and phase-transition mechanisms detailed in this report, the simulation manager will effectively orchestrate a non-equilibrium trophic cascade. The result will be a mathematically elegant, deeply biological, and visually profound generative art system capable of sustaining massive emergent complexity.

Works cited

1. Substrate evaporation drives collective construction in termites - eLife, accessed June 8, 2026, <https://elifesciences.org/reviewed-preprints/86843>
2. Metabolism during differentiation in the slime mold Physarum polycephalum, accessed June 8, 2026, <https://cdnsiencepub.com/doi/10.1139/m74-017>
3. Behavioural changes in slime moulds over time - PMC - NIH, accessed June 8, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC9939273/>
4. Termite gas emissions select for hydrogenotrophic microbial communities in termite mounds | PNAS, accessed June 8, 2026, <https://www.pnas.org/doi/10.1073/pnas.2102625118>

5. How termites ventilate - Harvard Gazette, accessed June 8, 2026, <https://news.harvard.edu/gazette/story/2015/09/how-termites-ventilate/>
6. Termite mounds harness diurnal temperature oscillations for ventilation - PNAS, accessed June 8, 2026, <https://www.pnas.org/doi/10.1073/pnas.1423242112>
7. The Emergence and Dynamical Evolution of Complex Transport Networks from Simple Low-Level Behaviours Author=Jeff Jones - arXiv, accessed June 8, 2026, <https://arxiv.org/pdf/1503.06579>
8. Solar-powered ventilation of African termite mounds. - SciSpace, accessed June 8, 2026, <https://scispace.com/pdf/solar-powered-ventilation-of-african-termite-mounds-2q8tfrd11c.pdf>
9. Characteristics of pattern formation and evolution in approximations of Physarum transport networks - PubMed, accessed June 8, 2026, <https://pubmed.ncbi.nlm.nih.gov/20067403/>
10. Newsletter Portfolio - 21/07/2025 - GitHub Pages, accessed June 8, 2026, <https://siasia-dev.github.io/portfolio-newsletter/latest.html>
11. Routing Physarum "Signals" with Chemicals | Request PDF - ResearchGate, accessed June 8, 2026, https://www.researchgate.net/publication/314606051_Routing_Physarum_Signals_with_Chemicals
12. Controllable Intrinsic Surface Pattern Generation Using Slime Mold Simulations, accessed June 8, 2026, https://www.researchgate.net/publication/403701535_Controllable_Intrinsic_Surface_Pattern_Generation_Using_Slime_Mold_Simulations
13. Icarus - Kevin Peter He, accessed June 8, 2026, <https://www.kevinpeterhe.com/icarus>
14. Generative Art Tools - SuperRare, accessed June 8, 2026, <https://superrare.com/curation/editorial/2145>
15. Adaptive behaviour and learning in slime moulds: the role of oscillations - PMC, accessed June 8, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC7935053/>
16. Random network peristalsis in Physarum polycephalum organizes fluid flows across an individual | PNAS, accessed June 8, 2026, <https://www.pnas.org/doi/10.1073/pnas.1305049110>
17. (PDF) Microplasmodium Dynamics of Physarum Polycephalum - ResearchGate, accessed June 8, 2026, https://www.researchgate.net/publication/251181953_Microplasmodium_Dynamics_of_Physarum_Polycephalum
18. Allometry in Physarum plasmodium during free locomotion: size versus shape, speed and rhythm - Company of Biologists journals, accessed June 8, 2026, <https://journals.biologists.com/jeb/article/218/23/3729/14430/Allometry-in-Physarum-plasmodium-during-free>
19. Caste-specific processing of construction pheromones in mound ..., accessed June 8, 2026, <https://www.biorxiv.org/content/10.1101/2025.09.08.674840v1.full-text>
20. Vibrational long-distance communication in the termites Macrotermes natalensis

- and *Odontotermes* sp. - Company of Biologists journals, accessed June 8, 2026, <https://journals.biologists.com/jeb/article/216/17/3249/11623/Vibrational-long-distance-communication-in-the>
21. THE TRAIL PHEROMONE OF THE TERMITE, TRINERVITERMES TRINERVOIDES - WALTER R. TSCHINKEL¹ and PETER G. CLOSE² - FSU Biology, accessed June 8, 2026, <https://www.bio.fsu.edu/~tschink/publications/1972-2.pdf>
 22. Modeling of communication within termite-like population using agent-based model: Pheromone evaporation effects - BIO Web of Conferences, accessed June 8, 2026, https://www.bio-conferences.org/articles/bioconf/pdf/2025/26/bioconf_istakcos2024_03006.pdf
 23. Trail Pheromones: An Integrative View of Their Role in Social Insect Colony Organization - Bristol Bee Group, accessed June 8, 2026, <https://www.socialinsect-research.com/resources/Publications/Czaczksetal.2015.pdf>
 24. Identification and evaluation of (3Z, 6Z, 8E)-dodeca-3, 6, 8-trien-1-ol as the trail following pheromone on *Microcerotermes exiguus* (Isoptera:Termitidae) - SciELO, accessed June 8, 2026, http://www.scielo.sa.cr/scielo.php?script=sci_arttext&pid=S0034-77442018000100303
 25. Validating a Termite-Inspired Construction Coordination Mechanism Using an Autonomous Robot - Frontiers, accessed June 8, 2026, <https://www.frontiersin.org/journals/robotics-and-ai/articles/10.3389/frobt.2021.645728/full>
 26. Excavation and aggregation as organizing factors in de novo ..., accessed June 8, 2026, <https://royalsocietypublishing.org/rspb/article/284/1856/20162730/78523/Excavation-and-aggregation-as-organizing-factors>
 27. Self-organized biotectonics of termite nests - PNAS, accessed June 8, 2026, <https://www.pnas.org/doi/10.1073/pnas.2006985118>
 28. Vibratory Alarm Signals in Two Sympatric Higher Termite Species: *Pseudacanthotermes spiniger* and *P. militaris* (Termitidae, Macrotermitinae) | Semantic Scholar, accessed June 8, 2026, <https://www.semanticscholar.org/paper/Vibratory-Alarm-Signals-in-Two-Sympatric-Higher-and-Conn%C3%A9table-Robert/24867b737822e8173048c0a6e67a7dfc2efe34f9>
 29. Vibrational alarm communication in the damp-wood termite *Zootermopsis nevadensis* | Request PDF - ResearchGate, accessed June 8, 2026, https://www.researchgate.net/publication/229962035_Vibrational_alarm_communication_in_the_damp-wood_termite_Zootermopsis_nevadensis
 30. Factors Impacting Aggression in *Reticulitermes* - Digital Showcase @ University of Lynchburg, accessed June 8, 2026, <https://digitalshowcase.lynchburg.edu/cgi/viewcontent.cgi?article=1306&context=utcp>
 31. The nature of alarm communication in *Constrictotermes cyphergaster* (Blattodea:

- Termitoidea: Termitidae): the integration of chemical and vibroacoustic signals - PMC, accessed June 8, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC4736033/>
32. Efficient flocking: metric versus topological interactions - Royal Society Publishing, accessed June 8, 2026, <https://royalsocietypublishing.org/rsos/article/8/9/202158/96185/Efficient-flocking-metric-versus-topological>
 33. Scale-free correlations in starling flocks - PMC, accessed June 8, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC2900681/>
 34. Diffusion and Topological Neighbours in Flocks of Starlings: Relating a Model to Empirical Data - PMC, accessed June 8, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC4436282/>
 35. University of Groningen Scale-free correlations, influential neighbours and speed control in flocks of birds Hemelrijk, Charlott - RUG, accessed June 8, 2026, https://pure.rug.nl/ws/files/17747148/Hemelrijk_2015_JStatPhys.pdf
 36. What it Takes to Fly: The Structural and Functional Respiratory Refinements in Birds and Bats - ResearchGate, accessed June 8, 2026, https://www.researchgate.net/publication/12319389_What_it_takes_to_fly_The_structural_and_functional_respiratory_refinements_in_birds_and_bats
 37. The physiological basis of bird flight - PMC - NIH, accessed June 8, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC4992708/>
 38. Metabolic reduction after long-duration flight is not related to fat-free mass loss or flight duration in a migratory passerine - NSF PAR, accessed June 8, 2026, <https://par.nsf.gov/servlets/purl/10217437>
 39. Metabolic scaling of fire ants (*Solenopsis invicta*) engaged in collective behaviors - PMC, accessed June 8, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC8905630/>
 40. Allometric Scaling of Metabolism, Growth, and Activity in Whole Colonies of the Seed-Harvester Ant *Pogonomyrmex californicus* - ResearchGate, accessed June 8, 2026, https://www.researchgate.net/publication/45826621_Allometric_Scaling_of_Metabolism_Growth_and_Activity_in_Whole_Colonies_of_the_Seed-Harvester_Ant_Pogonomyrmex_californicus
 41. Respiratory concerts revealed by scanning microrespirography in a termite *Prorhinotermes simplex* (Isoptera: Rhinotermitidae) - PubMed, accessed June 8, 2026, <https://pubmed.ncbi.nlm.nih.gov/17306293/>
 42. Effects of sinusoidal 60-Hz electric and magnetic fields on ATP and oxygen levels in the slime mold, *Physarum polycephalum* - ResearchGate, accessed June 8, 2026, https://www.researchgate.net/publication/19407469_Effects_of_sinusoidal_60-Hz_electric_and_magnetic_fields_on_ATP_and_oxygen_levels_in_the_slime_mold_Phy_sarum_polycephalum
 43. Elf (extremely low frequency) communications system ecological monitoring program. The effects of exposing the slime mold *Physarum polycephalum* to electromagnetic fields (Technical Report) | OSTI.GOV, accessed June 8, 2026, <https://www.osti.gov/biblio/5638440>

44. (PDF) Limitations of the Q10 Coefficient for Quantifying Temperature Sensitivity of Anaerobic Organic Matter Decomposition: A Modeling Based Assessment - ResearchGate, accessed June 8, 2026,
https://www.researchgate.net/publication/353708552_Limitations_of_the_Q10_Coefficient_for_Quantifying_Temperature_Sensitivity_of_Anaerobic_Organic_Matter_Decomposition_A_Modeling_Based_Assessment
45. accessed June 8, 2026,
<https://www.ovid.com/journals/frfcn/pdf/10.1007/s11461-006-0018-4~characteristics-of-the-temperature-coefficientq10-for-the#:~:text=For%20soil%20respiration%2C%20the%20median,range%20from%201.0%20to%204.0.>
46. Characteristics of the temperature coefficient, Q10 - Ovid, accessed June 8, 2026,
<https://www.ovid.com/journals/frfcn/pdf/10.1007/s11461-006-0018-4~characteristics-of-the-temperature-coefficientq10-for-the>
47. Termite mounds harness diurnal temperature oscillations for ventilation - Soft Math Lab, accessed June 8, 2026,
<https://softmath.seas.harvard.edu/wp-content/uploads/2019/10/2015-12.pdf>
48. Metabolic adjustments to increasing foraging costs of starlings in a closed economy, accessed June 8, 2026,
<https://journals.biologists.com/jeb/article/208/21/4099/15842/Metabolic-adjustments-to-increasing-foraging-costs>